

# Documentation

A guide to using L<sup>A</sup>T<sub>E</sub>X File Template v.4.0.0.

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# 1 Introduction

## Abstract

Thank you for choosing to use L<sup>A</sup>T<sub>E</sub>X File Template. If you find any bugs or want some feature not currently included, then my contact information can be found on my GitHub: <https://github.com/MatthewEmer>.

This template is intended for academics (with special focus on those in the fields of computer science or mathematics) who want to write papers or other related documents in a consistent format which ties in with standard practises from other published papers.

This guide consists of setup instructions to download the files and create your first document, before moving onto the custom commands/environments designed to speed up academic writing. Before using this guide, I would suggest that you have some practise with some basic L<sup>A</sup>T<sub>E</sub>X commands using a guide such as <https://www.overleaf.com/learn>.

This documentation is accurate as of Version 4.0.0.

## Acknowledgements

I would like to thank Dr Jamie Mason for his support and expertise in leading my discrete mathematics tutorials and seminars this past year, without which this project would probably not exist. His continuous feedback throughout has allowed me to hone the project from the messy original versions into a useable tool.

Thanks also to Joseph Eddon for supporting the development of this template through the creation of his own L<sup>A</sup>T<sub>E</sub>X template, and for the feedback and suggestions he provided on multiple versions of the template throughout last year.

## 2 Document Setup

### 2.1 Project Files

The actual L<sup>A</sup>T<sub>E</sub>X file template is contained in the folder *The Template* and consists of four files: *template.tex*, *\_documentContent.tex*, *\_appendix.tex*, and *template.pdf*. To use the template, all four files must be downloaded and placed in the same folder.

#### 2.1.1 \_appendix.tex

This file allows you to add an optional appendix to the end of your document which is not counted towards your page count. By default it contains a unnumbered part title which has been inserted into the table of contents.

#### 2.1.2 \_documentContent.tex

This file allows you to add all the main content of the document you are trying to write. It will be inserted after the table of contents, but before the appendix (if you choose to add one). By default it contains an example section and definition. Only pages in this file are counted towards the page count of the document.

#### 2.1.3 template.pdf

This file contains the output of the L<sup>A</sup>T<sub>E</sub>X compiler. If you want to rename this document while still working on it, then delete this version, alongside any build files your compiler has created and change the name of *template.tex*.

#### 2.1.4 template.tex

This file sets up the structure of the document, as well as acting as a compiler for the other two *.tex* files, ensuring they get inserted into the right sections of the document. When creating your document, you need to edit lines 12-15 of this file to create the title page, alongside the document header and footer, as described in Table 1.

| Line | Variable Name | Description                                                                  |
|------|---------------|------------------------------------------------------------------------------|
| 12   | thetitle      | The document's title. This will be placed in the header.                     |
| 13   | thesubtitle   | The document's (optional) subtitle. This will be placed in the header.       |
| 14   | theauthor     | The list of authors. This will be placed in the footer.                      |
| 15   | thedata       | Replace the code if you want a static date rather than date of last compile. |

Table 1: Lines to edit.

### 2.2 Optional Files

#### 2.2.1 Storing Images

If you choose to include figures in your project, then you will need to create a subfolder labelled *Images* where you can place the image files for the project.

#### 2.2.2 GitHub Repositories

If you are placing the document within a GitHub repository, you may want to copy over the *.gitignore* from this repository as it is set up to ignore any L<sup>A</sup>T<sub>E</sub>X build files generated by LaTeX Workshop on Visual Studio Code. These are: *\*.aux*, *\*.fdb\_latexmk*, *\*.fls*, *\*.log*, *\*.out*, *\*.gx*, *\*.toc*, and *\*.gz*.

## 3 Custom Commands

### 3.1 General Tools

#### 3.1.1 Element Spacing

**Command 3.1.1.1:** `\separate`

The `separate` command inserts vertical spacing between two elements whilst simultaneously removing the new paragraph indent from the following element. You can specify the amount of spacing as described in Table 2 and as seen in Example 3.1.1.1.

| No. | Name    | Description                               | Example Value |
|-----|---------|-------------------------------------------|---------------|
| 1   | Spacing | The number of points of vertical spacing. | 5             |

Table 2: Parameters for `separate`.

**Example 3.1.1.1:** `\separate{3}` produces the separation as seen between the Command line and the description of `separate`.

#### 3.1.2 Text Formatting

**Command 3.1.2.1:** `\colouredText`

The `colouredText` command takes the given text and outputs it in a different colour, as chosen by the writer. This can be seen in Example 3.1.2.1 and as described in Table 3.

| No. | Name   | Description                            | Example Value |
|-----|--------|----------------------------------------|---------------|
| 1   | Colour | The colour to change the text to.      | blue          |
| 2   | Text   | The text you want outputted in colour. | Hello World   |

Table 3: Parameters for `colouredText`.

**Example 3.1.2.1:** `\colouredText{blue}{Hello World}` produces **Hello World**.

**Command 3.1.2.2:** `\comment`

The `comment` command inserts notes into the document in red so they can be found more easily. This can be seen in Example 3.1.2.2 and as described in Table 4.

| No. | Name | Description                            | Example Value |
|-----|------|----------------------------------------|---------------|
| 1   | Text | The text you want outputted in colour. | Hello World   |

Table 4: Parameters for `comment`.

**Example 3.1.2.2:** `\comment{Hello World}` produces **Hello World**.

#### 3.1.3 Date Formatting

**Command 3.1.3.1:** `\yeardate`

The `yeardate` command takes an adjacent date variable and reformats it into the four-digit year before outputting to the page as seen in Example 3.1.3.1.

| No. | Name | Description | Example Value |
|-----|------|-------------|---------------|
|     | n/a  |             |               |

Table 5: Parameters for `yeardate`.

**Example 3.1.3.1:** `\date{\yeardate\today}` produces 2026.

**Command 3.1.3.2:** `\monthyeardate`

The monthyeardate command takes an adjacent date variable and reformats it into the full month and four-digit year before outputting to the page as seen in Example 3.1.3.2.

| No. | Name | Description | Example Value |
|-----|------|-------------|---------------|
|     | n/a  |             |               |

Table 6: Parameters for monthyeardate.

**Example 3.1.3.2:** `\date{\monthyeardate\today}` produces August 2026.

## 3.2 Tables and Figures

### 3.2.1 Inserting Tables

**Command 3.2.1.1:** `\inserttable`

The inserttable command simplifies the process of inserting tables into your document as described in Table 7 and seen in Example 3.2.1.1.

| No. | Name    | Description                            | Example Value                    |
|-----|---------|----------------------------------------|----------------------------------|
| 1   | Columns | The required columns, separated by  s. | c c c c                          |
| 2   | Caption | The caption to appear under the table. | Parameters for inserting tables. |
| 3   | Headers | The headers, separated by ampersands.  | No. & Name & ...                 |
| 4   | Content | The content to be placed in the table. |                                  |

Table 7: Parameters for inserttable.

**Example 3.2.1.1:** `\inserttable{c|c|c|c}{Parameters...}{No. & Name & ...}{1 & Columns & ...}` produces Table 7.

### 3.2.2 Inserting Figures

**Command 3.2.2.1:** `\insertimage`

The insertimage command simplifies the process of inserting figures into your document as described in Table 8 and seen in Example 3.2.2.1.

| No. | Name           | Description                                             | Example Value     |
|-----|----------------|---------------------------------------------------------|-------------------|
| 1   | File name      | The file name of the image (must be in <i>Images</i> .) | durham.jpg        |
| 2   | Figure caption | The caption to appear under the figure.                 | Durham Cathedral. |
| 3   | Width          | The amount of the page width the image should occupy.   | 0.3               |

Table 8: Parameters for insertimage.

**Example 3.2.2.1:** `\insertimage{durham.jpg}{A large building sitting on top of a lush green hillside. (Stoll, 2022)}{0.3}` produces Figure 1.

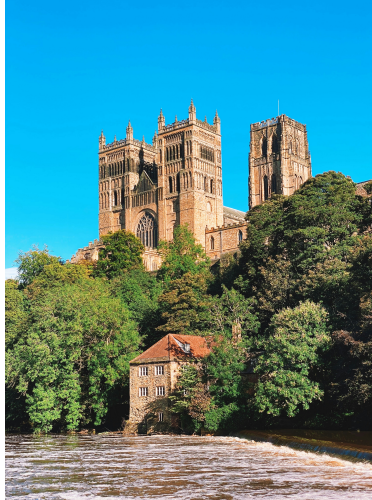


Figure 1: A large building sitting on top of a lush green hillside. (Stoll, 2022)

### 3.2.3 Referencing Tables and Figures

**Command 3.2.3.1:** `\elemref`

The `elemref` command allows you to easily reference tables and figures with clickable links. This is described in Table 9 and as seen in Example 3.2.3.1.

| No. | Name  | Description                               | Example Value               |
|-----|-------|-------------------------------------------|-----------------------------|
| 1   | Label | The label of the element being linked to. | tab:Parameters for elemref. |
| 2   | Type  | The type of element being linked to.      | Table                       |

Table 9: Parameters for `elemref`.

**Example 3.2.3.1:** `\elemref{tab:Parameters for elemref.}{Table}` produces the reference Table 9.

## 3.3 Academic Environments

### 3.3.1 Basic Environments

**Command 3.3.1.1:** `\definition`

The `definition` command creates an environment for setting out a definition with automatic numbering with inbuilt labels so that you can reference the definition later in the document. Its parameters can be seen in Table 10 and an example of its use can be seen in Example 3.3.1.1.

| No. | Name       | Description                                                   | Example Value           |
|-----|------------|---------------------------------------------------------------|-------------------------|
| 1   | Label      | The word being defined so it can be easily referenced.        | example                 |
| 2   | Definition | The actual definition of the word (with the word underlined). | An <u>example</u> is... |

Table 10: Parameters for `definition`.

**Example 3.3.1.1:** `\definition{example}{An \underline{example} is what this is.}` produces Definition 3.3.1.1

**Definition 3.3.1.1:** An example is what this is.

**Command 3.3.1.2:** `\conjecture`

The conjecture command creates an environment for setting out a conjecture with automatic numbering with inbuilt labels so that you can reference the conjecture later in the document. Its parameters can be seen in Table 11 and an example of its use can be seen in Example 3.3.1.2.

| No. | Name       | Description                     | Example Value                  |
|-----|------------|---------------------------------|--------------------------------|
| 1   | Label      | The conjecture's name.          | Someone's Conjecture           |
| 2   | Conjecture | The content for the conjecture. | Someone's Conjecture states... |

Table 11: Parameters for conjecture.

**Example 3.3.1.2:** `\conjecture{Someone's Conjecture}{Someone's Conjecture states something or other.}` produces Conjecture 3.3.1.1

**Conjecture 3.3.1.1:** Someone's Conjecture states something or other.

**3.3.2 Environments with Proofs or Solutions****Command 3.3.2.1:** `\theorem`

The theorem command creates an environment for setting out a theorem with automatic numbering with inbuilt labels so that you can reference the theorem later in the document. Its parameters can be seen in Table 12 and an example of its use can be seen in Example 3.3.2.1.

| No. | Name       | Description               | Example Value                |
|-----|------------|---------------------------|------------------------------|
| 1   | Label      | The theorem's name.       | Example Theorem              |
| 2   | Definition | The theorem definition.   | If $2x = y$ then $8x = 4y$ . |
| 3   | Proof      | The proof of the theorem. | If $2x = y$ ...              |

Table 12: Parameters for theorem.

**Example 3.3.2.1:** `\theorem{Example Theorem}{If  $2x = y$  then...}{If  $2x = y$ , we...}` produces Theorem 3.3.2.1

**Theorem 3.3.2.1:** If  $2x = y$  then  $8x = 4y$  holds for all  $(x, y) \in \mathbb{N}^2$ .

*Proof:* If  $2x = y$ , we can factor  $8x = 4y$  by taking out a common multiplier of 4 to get  $4(2x = y)$ . Therefore, the theorem must hold for all  $(x, y) \in \mathbb{N}^2$ .

□

**Command 3.3.2.2:** `\lemma`

The lemma command creates an environment for setting out a lemma with automatic numbering with inbuilt labels so that you can reference the lemma later in the document. Its parameters can be seen in Table 13 and an example of its use can be seen in Example 3.3.2.2.

| No. | Name       | Description             | Example Value                |
|-----|------------|-------------------------|------------------------------|
| 1   | Label      | The lemma's name.       | Example Lemma                |
| 2   | Definition | The lemma definition.   | If $2x = y$ then $8x = 4y$ . |
| 3   | Proof      | The proof of the lemma. | If $2x = y$ ...              |

Table 13: Parameters for lemma.

**Example 3.3.2.2:** `\lemma{Example Lemma}{If  $2x = y$  then...}{If  $2x = y$ , we...}` produces Lemma 3.3.2.1

**Lemma 3.3.2.1:** If  $2x = y$  then  $8x = 4y$  holds for all  $(x, y) \in \mathbb{N}^2$ .

*Proof:* If  $2x = y$ , we can factor  $8x = 4y$  by taking out a common multiplier of 4 to get  $4(2x = y)$ . Therefore, the lemma must hold for all  $(x, y) \in \mathbb{N}^2$ . □

**Command 3.3.2.3:** `\proposition`

The proposition command creates an environment for setting out a proposition with automatic numbering with inbuilt labels so that you can reference the proposition later in the document. Its parameters can be seen in Table 14 and an example of its use can be seen in Example 3.3.2.3.

| No. | Name       | Description                   | Example Value                |
|-----|------------|-------------------------------|------------------------------|
| 1   | Label      | The proposition's name.       | Example Proposition          |
| 2   | Definition | The proposition definition.   | If $2x = y$ then $8x = 4y$ . |
| 3   | Proof      | The proof of the proposition. | If $2x = y$ ...              |

Table 14: Parameters for proposition.

**Example 3.3.2.3:** `\proposition{Example Proposition}{If  $2x = y$  then...}{If  $2x = y$ , we...}` creates Proposition 3.3.2.1

**Proposition 3.3.2.1:** If  $2x = y$  then  $8x = 4y$  holds for all  $(x, y) \in \mathbb{N}^2$ .

*Proof:* If  $2x = y$ , we can factor  $8x = 4y$  by taking out a common multiplier of 4 to get  $4(2x = y)$ . Therefore, the lemma must hold for all  $(x, y) \in \mathbb{N}^2$ . □

**Command 3.3.2.4:** `\standardproblem`

The standardproblem command creates an environment for setting out a standard problem with inbuilt labels so that you can reference the standard problem later in the document. Its parameters can be seen in Table 15 and an example of its use can be seen in Example 3.3.2.4. When referencing a standard problem, the native `\hyperref` command should be used.

| No. | Name     | Description             | Example Value               |
|-----|----------|-------------------------|-----------------------------|
| 1   | Number   | The problem number.     | 1                           |
| 2   | Problem  | The problem statement.  | If $2x = 4$ , what is $x$ ? |
| 3   | Solution | The problem's solution. | $2x = 4$ , divide...        |

Table 15: Parameters for standardproblem.

**Example 3.3.2.4:** `\standardproblem{1}{If  $2x = 4$  what is  $x$ }{ $2x = 4$ ...}` creates Standard Problem #1.

**Standard Problem #1:** If  $2x = 4$ , what is  $x$ ?

*Solution:*  $2x = 4$ , divide both sides by 2,  $x = 2$ . □

### 3.3.3 Sub-Environments

**Important Note:** Both of the following commands can only be referenced using `\elemref` as opposed to the environment references used by the rest of this section.

**Command 3.3.3.1:** `\proof`

The proof command creates an environment for setting out a proof with inbuilt labels so that you can reference the proof later in the document. Its parameters can be seen in Table 16 and an example of its use can be seen in Example 3.3.3.1.

| No. | Name  | Description                | Example Value   |
|-----|-------|----------------------------|-----------------|
| 1   | Label | The proof's name.          | Example Proof   |
| 2   | Proof | The content for the proof. | Let $x = \dots$ |

Table 16: Parameters for proof.

**Example 3.3.3.1:** `\proof{prop:Example Proposition}{If  $2x = y\dots$ }` produces Proof 3.3.2.

**Command 3.3.3.2:** `\solution`

The solution command creates an environment for setting out a solution with inbuilt labels so that you can reference the solution later in the document. Its parameters can be seen in Table 17 and an example of its use can be seen in Example 3.3.3.2.

| No. | Name     | Description                   | Example Value    |
|-----|----------|-------------------------------|------------------|
| 1   | Label    | The solution's name.          | Example Solution |
| 2   | Solution | The content for the solution. | Let $x = \dots$  |

Table 17: Parameters for solution.

**Example 3.3.3.2:** `\solution{1}{ $2x = 4$ , divide...}` produces Solution 3.3.2.

**3.3.4 Customisable Environments****Command 3.3.4.1:** `\numberedentry`

The numberedentry command creates an environment for setting out a custom environment with automatic numbering with inbuilt labels so that you can reference it later in the document. Its parameters can be seen in Table 18 and an example of its use can be seen in Example 3.3.4.1.

| No. | Name       | Description               | Example Value                |
|-----|------------|---------------------------|------------------------------|
| 1   | Label      | The entry's name.         | cmd:numberedentry            |
| 2   | Entry Type | The type of entry.        | Command                      |
| 3   | Content    | The content of the entry. | The numberedentry command... |

Table 18: Parameters for numberedentry.

**Example 3.3.4.1:** `\numberedentry` produces Command 3.3.4.1.

**3.3.5 Referencing Environments****Command 3.3.5.1:** `\preenvref`

The preenvref command allows you to easily reference academic environments with clickable links before they are defined in the document. This is described in Table 19 and as seen in Example 3.3.5.1.

| No. | Name    | Description                                   | Example Value     |
|-----|---------|-----------------------------------------------|-------------------|
| 1   | Label   | The label of the environment being linked to. | ex:cmd:postenvref |
| 2   | Type    | The type of environment being linked to.      | Example           |
| 3   | Counter | The name of the counter for that environment. | examplenummer     |

Table 19: Parameters for preenvref and postenvref.

**Example 3.3.5.1:** `\preenvref{ex:cmd:preenvref}{Example}{examplenum}` produces the reference Example 3.3.5.1.

**Command 3.3.5.2:** `\postenvref`

The `postenvref` command allows you to easily reference academic environments with clickable links after they are defined in the document. This is described in Table 19 and as seen in Example 3.3.5.2.

**Example 3.3.5.2:** `\postenvref{ex:cmd:preenvref}{Example}{examplenum}` produces the reference Example 3.3.5.1.

## 3.4 Mathematics

### 3.4.1 Basic Mathematics Notation

**Command 3.4.1.1:** Insert standard number set symbols.

This set of commands allows you to quickly insert the set symbols for all of the common number sets used in mathematics. The supported sets, and their associated commands, can be seen in Table 20 below.

| Number Set | Command         | Output       |
|------------|-----------------|--------------|
| Natural    | <code>\N</code> | $\mathbb{N}$ |
| Integers   | <code>\Z</code> | $\mathbb{Z}$ |
| Rational   | <code>\Q</code> | $\mathbb{Q}$ |
| Real       | <code>\R</code> | $\mathbb{R}$ |
| Imaginary  | <code>\I</code> | $\mathbb{I}$ |
| Complex    | <code>\C</code> | $\mathbb{C}$ |

Table 20: Supported number sets.

### 3.4.2 Vectors and Matrices

**Command 3.4.2.1:** `\dotproduct`

The `dotproduct` command allows you to easily use the dot product vector operation within equations. The parameters for the function are described in Table 21, and an example can be seen Example 3.4.2.1.

| No. | Name     | Description                              | Example Value |
|-----|----------|------------------------------------------|---------------|
| 1   | Vector A | The vector on the left of the operation. | $\mathbf{u}$  |
| 2   | Vector B | The vector on the right of the operation | $\mathbf{v}$  |

Table 21: Parameters for `dotproduct`.

**Example 3.4.2.1:** `\dotproduct{u}{v}` produces  $\langle \mathbf{u}, \mathbf{v} \rangle$ .

**Command 3.4.2.2:** `\jacobian`

The `Jacobian` command inserts the notation for the Jacobian matrix of a given function. This is described in Table 22, and can be seen in Example 3.4.2.2.

| No. | Name     | Description                                  | Example Value |
|-----|----------|----------------------------------------------|---------------|
| 1   | Function | The function the Jacobian is generated from. | $\mathbf{f}$  |

Table 22: Parameters for `jacobian`.

**Example 3.4.2.2:** `\jacobian{f}` produces  $\mathbf{J}_f$ .

**Command 3.4.2.3:** `\hessian`

The Hessian command inserts the notation for the Hessian matrix of a given function. This is described in Table 23, and can be seen in Example 3.4.2.3.

| No. | Name     | Description                                 | Example Value |
|-----|----------|---------------------------------------------|---------------|
| 1   | Function | The function the Hessian is generated from. | f             |

Table 23: Parameters for hessian.

**Example 3.4.2.3:** `\hessian{f}` produces  $\mathbf{H}_f$ .

**Command 3.4.2.4:** `\lagrangian`

The Lagrangian command inserts the notation for a Lagrangian function. This is described in Table 24, and can be seen in Example 3.4.2.4.

| No. | Name | Description | Example Value |
|-----|------|-------------|---------------|
|     | n/a  |             |               |

Table 24: Parameters for lagrangian.

**Example 3.4.2.4:** `\lagrangian` produces  $\mathcal{L}$ .

## 3.5 Computer Science

### 3.5.1 Relational Algebra

**Command 3.5.1.1:** Insert row join symbols.

This set of commands allows you to quickly insert the symbols for all of the common join types used in relational algebra. The supported joins, and their associated commands, can be seen in Table 25 below.

| Number Set       | Command                      | Output    |
|------------------|------------------------------|-----------|
| Inner Join       | <code>\innerjoin</code>      | $\bowtie$ |
| Left Outer Join  | <code>\leftouterjoin</code>  | $\Join$   |
| Right Outer Join | <code>\rightouterjoin</code> | $\Join$   |
| Full Outer Join  | <code>\fullouterjoin</code>  | $\Join$   |

Table 25: Supported joins.

### 3.5.2 Algorithms

**Command 3.5.2.1:** `\insertalgorithm`

The insertalgorithm command inserts a box suited for a pseudocode algorithm to be inserted into the document. This is described in Table 26, and can be seen in Example 3.5.2.1.

| No. | Name           | Description                      | Example Value                                |
|-----|----------------|----------------------------------|----------------------------------------------|
| 1   | Algorithm Name | The name of the algorithm.       | Hello or goodbye                             |
| 2   | Inputs         | The required inputs.             | n/a                                          |
| 3   | Outputs        | Any returned values.             | n/a                                          |
| 4   | Pseudocode     | The pseudocode of the algorithm. | <code>\Procedure{helloOrGoodbye}{}...</code> |

Table 26: Parameters for insertalgorithm.

**Example 3.5.2.1:** `\insertalgorithm{Hello or goodbye}{n/a}{n/a}{\Procedure{helloOrGoodbye}{...}}` produces Algorithm 1.

---

**Algorithm 1** Hello or goodbye

---

**Input:**  $x \in \{1, 2\}$

**Output:** A string message.

```
procedure HELLOORGODBYE(x)
  if (x == 1) then return "Hello World"
  else if (x == 2) then return "Goodbye"
  else return "418 I am a teapot."
  end if
end procedure
```

---

## 4 Appendix

### 4.1 List of Figures

#### Figure 1

Stoll, M. (2022). A large building sitting on top of a lush green hillside. [Online] unsplash.com.  
Available at: <https://unsplash.com/photos/a-large-building-sitting-on-top-of-a-lush-green-hillside-jATW0-hO4-I> [Accessed 10 Aug. 2026].